

Answer Key: Chapter 8 - Predicting What Comes Next

Exercise Set 8.1

1. Find the first five terms:

- (i) $t_n = 3n - 4$

Terms: -1, 2, 5, 8, 11

- (ii) $t_n = 2 - 5n$

Terms: -3, -8, -13, -18, -23

- (iii) $t_n = n^2 - 2n + 3$

Terms: 2, 3, 6, 11, 18

2. Find the 10th and 15th terms of $t_n = 5n - 3$:

- $t_{10} = 5(10) - 3 = 47$

- $t_{15} = 5(15) - 3 = 72$

3. Determine whether 97 and 172 are terms of $t_n = 5n - 3$:

- For 97: $5n - 3 = 97 \Rightarrow 5n = 100 \Rightarrow n = 20$ (Yes, 20th term)

- For 172: $5n - 3 = 172 \Rightarrow 5n = 175 \Rightarrow n = 35$ (Yes, 35th term)

4. Which term of $t_n = 5n - 3$ is 607?

- $5n - 3 = 607 \Rightarrow 5n = 610 \Rightarrow n = 122$ (122nd term)

5. Given $t_1 = -5$, $t_{n+1} = t_n + 3$:

- First 5 terms: -5, -2, 1, 4, 7

- Explicit rule: $t_n = 3n - 8$

- $3n - 8 = 52 \Rightarrow 3n = 60 \Rightarrow n = 20$ (Yes, 52 is the 20th term)

6. Given $T_1=1$, $T_2=2$, $T_3=4$, and $T_n = T_{n-1} + T_{n-2} + T_{n-3}$:

- $T_4 = 4 + 2 + 1 = 7$
- $T_5 = 7 + 4 + 2 = 13$
- $T_6 = 13 + 7 + 4 = 24$
- $T_7 = 24 + 13 + 7 = 44$
- $T_8 = 44 + 24 + 13 = 81$

Exercise Set 8.2

1. AP: 3, 8, 13, 18, ...

- $t_{10} = 3 + 9(5) = 48$
- $t_{26} = 3 + 25(5) = 128$

2. AP: 21, 18, 15, ...

- For -81: $21 + (n - 1)(-3) = -81 \Rightarrow n = 35$ (35th term)
- For 0: $21 + (n - 1)(-3) = 0 \Rightarrow n = 8$ (Yes, 0 is the 8th term)

3. AP: 11, 8, 5, 2, ...

- Explicit rule: $t_n = 14 - 3n$
- Recursive rule: $t_1 = 11$, $t_n = t_{n-1} - 3$ (for $n \geq 2$)

4. AP with 50 terms, $t_3 = 12$, $t_{50} = 106$:

- $a + 2d = 12$ and $a + 49d = 106 \Rightarrow d = 2$, $a = 8$
- $t_{29} = 8 + 28(2) = 64$

5. 2-digit numbers divisible by 3:

- Sequence: 12, 15, ..., 99 $\Rightarrow$ Number of terms (n) = 30
- Sum (S_{30}) = $(30 / 2) \times (12 + 99) = 1665$

6. Salary AP ($a = ₹5,00,000$, $d = ₹20,000$):

- $500000 + (n - 1)20000 = 700000 \Rightarrow n = 11$
- Income reaches ₹7,00,000 after 10 years (in the 11th year).

7. Marbles in rows 1 to 25:

- Total marbles (S_{25}) = $(25 \times 26) / 2 = 325$

Exercise Set 8.3

1. GP with $r = 2$ and $t_8 = 192$:

- $a \times 2^7 = 192 \Rightarrow a = 1.5$
- $t_{12} = 1.5 \times 2^{11} = 3072$

2. GP: 5, 25, 125, ...

- $t_n = 5^n$
- $t_{10} = 5^{10} = 9,765,625$

3. $t_1 = 2$, $t_{n+1} = 3t_n - 2$:

- Sequence: 2, 4, 10, 28, 82, 244, 730
- 730 is the 7th term (t_7).

4. GP: 2, 6, 18, ...

- Explicit formula: $t_n = 2 \cdot 3^{n-1}$
- Recursive formula: $t_1 = 2$, $t_n = 3t_{n-1}$ (for $n \geq 2$)
- $2 \cdot 3^{n-1} = 4374 \Rightarrow n = 8$ (8th term)

5. Bouncing ball (Initial height = 80 m, rebound ratio = 0.6):

- (i) Height after 5th bounce = $80 \times (0.6)^5 = 6.2208$ m
- (ii) Total vertical distance until 6th hit = $80 + 2 \times [\text{Sum of first 5 bounces}] = 301.3376$ m

6. GP: 2, $2\sqrt{2}$, 4, ...

- $a = 2$, $r = \sqrt{2}$
- $2 \cdot (\sqrt{2})^{n-1} = 128 \Rightarrow n = 13$ (13th term)

7. Sierpiński square carpet:

- (i) Red squares: Stage 0 = 1, Stage 1 = 8, Stage 2 = 64, Stage 3 = 512

- (ii) Stage 4 = 4096, Stage 5 = 32,768
- (iii) Explicit: $t_n = 8^n$ | Recursive: $t_0 = 1, t_n = 8t_{n-1}$
- (iv) Area formula: $A_n = (8/9)^n$ | As $n \rightarrow \infty$, area approaches 0.

End-of-Chapter Exercises

1. AP with $t_{11} = 38, t_{16} = 73$:

- $d = 7, a = -32$
- $t_{31} = -32 + 30(7) = 178$

2. AP with $t_3 = 16$ and $t_7 = t_5 + 12$:

- $d = 6, a = 4$
- AP: 4, 10, 16, 22, 28, ...

3. 3-digit numbers divisible by 7:

- AP: 105, 112, ..., 994 $\Rightarrow$ Number of terms (n) = 128

4. Multiples of 4 between 10 and 250:

- AP: 12, 16, ..., 248 $\Rightarrow$ Number of terms (n) = 60

5. GP with $a + ar = 4$ and $t_5 = 4t_3$:

- If $r = 2 \Rightarrow a = 4/3$. GP: $4/3, 8/3, 16/3, \dots$
- If $r = -2 \Rightarrow a = -4$. GP: $-4, 8, -16, 32, \dots$

6. Expressing 100 as sum of consecutive natural numbers:

- 5 terms: $18 + 19 + 20 + 21 + 22 = 100$
- 8 terms: $9 + 10 + 11 + 12 + 13 + 14 + 15 + 16 = 100$

7. Bacteria culture doubling every hour (Initial = 30):

- End of 2nd hour: 120
- End of 4th hour: 480
- End of n^{th} hour: 30×2^n

8. AP with $t_4 + t_8 = 24$ and $t_6 + t_{10} = 44$:

- $d = 5, a = -13$
- First 3 terms: -13, -8, -3

9. Smallest n such that $n(n+1)/2 > 1000$:

- $n = 45$ (Sum $S_{45} = 1035$)

10. GP: 2, 8, 32, ...

- Explicit: $t_n = 2 \cdot 4^{n-1}$ | Recursive: $t_1 = 2, t_n = 4t_{n-1}$
- $2 \cdot 4^{n-1} = 131072 \Rightarrow n = 9$ (9th term)

11. GP 3 terms sum = 13/12, product = -1:

- Terms: $3/4, -1, 4/3$ OR $4/3, -1, 3/4$

12. Prove x, y, z are in GP if $t_4=x, t_{10}=y, t_{16}=z$:

- $x = ar^3, y = ar^9, z = ar^{15}$
- $y^2 = a^2r^{18}$ and $xz = (ar^3)(ar^{15}) = a^2r^{18}$
- Since $y^2 = xz$, x, y, z form a GP.

13. GP 3 terms sum = 26, sum of squares = 364:

- Terms: 2, 6, 18 (or 18, 6, 2)

14. $P_1 = 1, P_2 = 2, P_n = P_1 + \dots + P_{n-1} + 1$:

- Values P_1 to P_8 : 1, 2, 4, 8, 16, 32, 64, 128
- Simpler recursive formula: $P_1 = 1, P_2 = 2, P_n = 2P_{n-1}$ (for $n \geq 3$)
- Explicit formula: $P_n = 2^{n-1}$

15. $W_1 = 1, W_2 = 2, W_n = W_1 + \dots + W_{n-2} + 2$:

- Values W_1 to W_8 : 1, 2, 3, 5, 8, 13, 21, 34
- Sequence: Virahānka-Fibonacci sequence